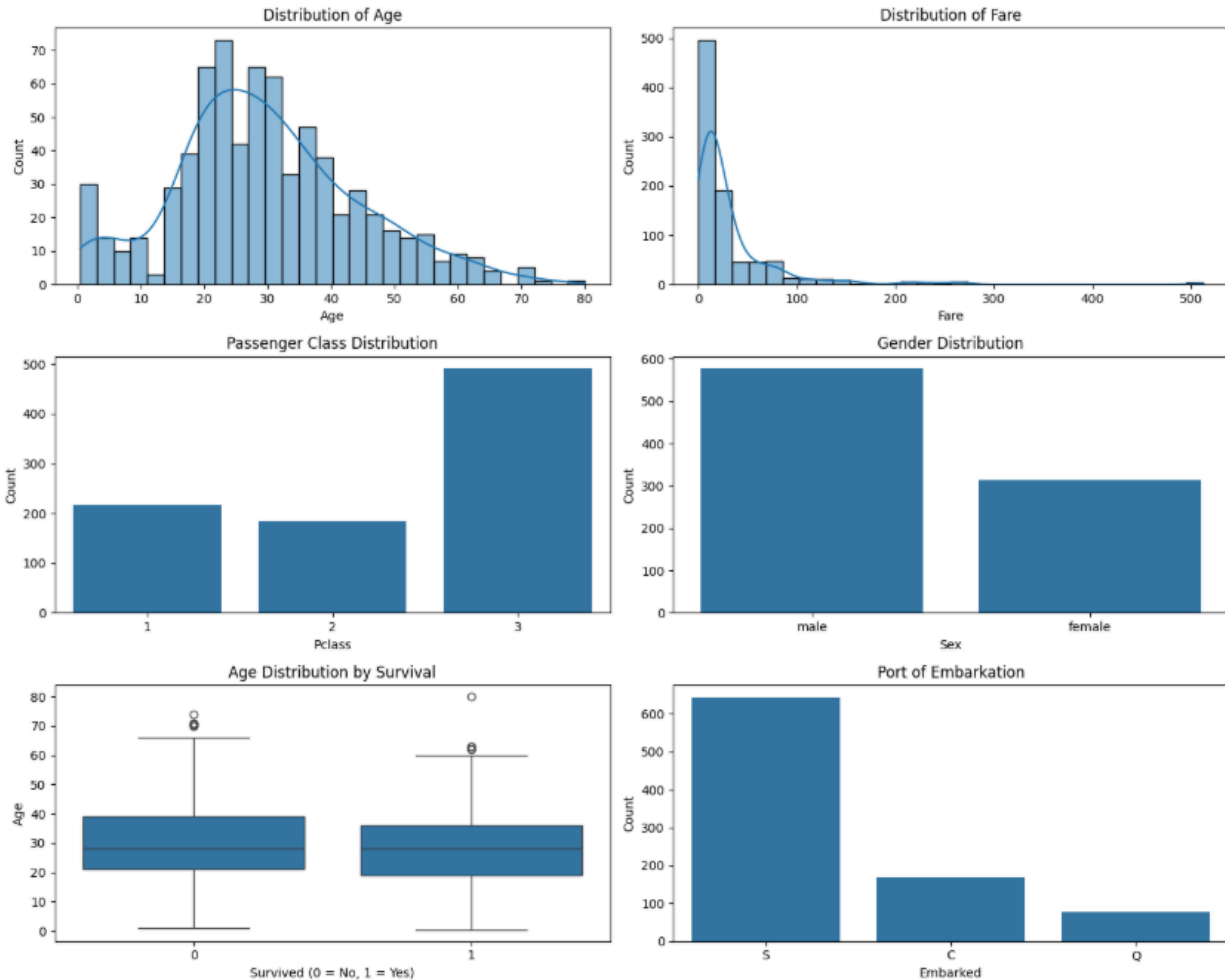


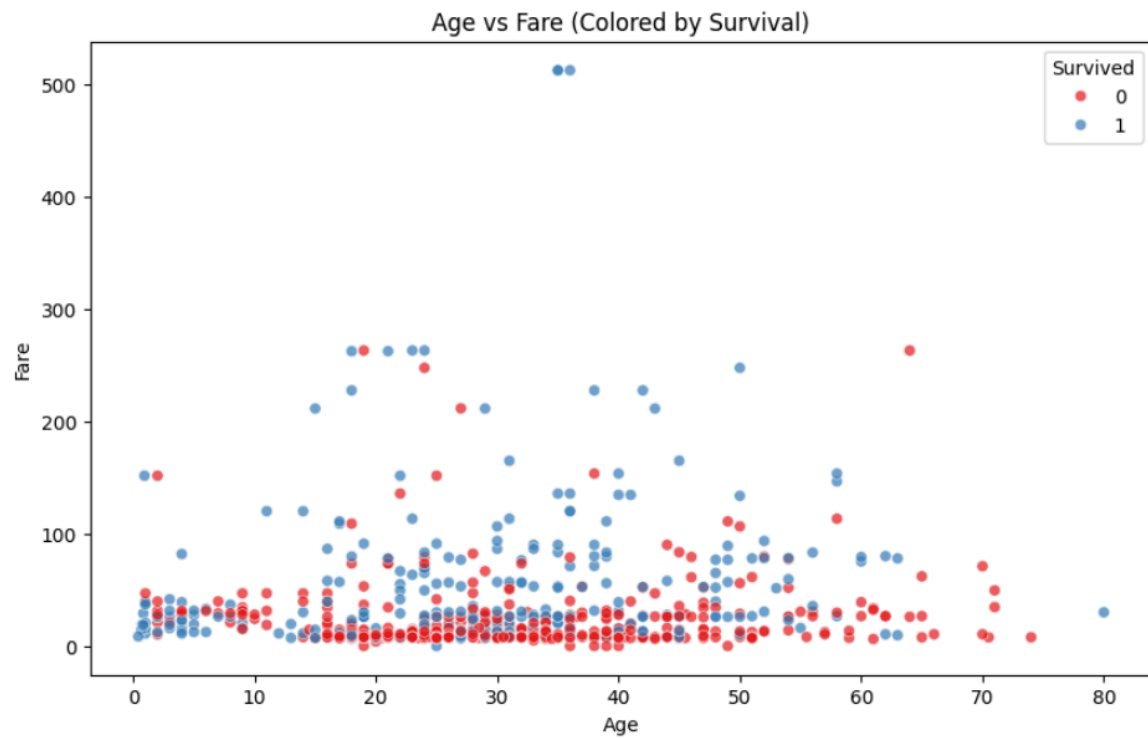
Train Dataset Summary (EDA) - Task5

Univariate Analysis



Observations:

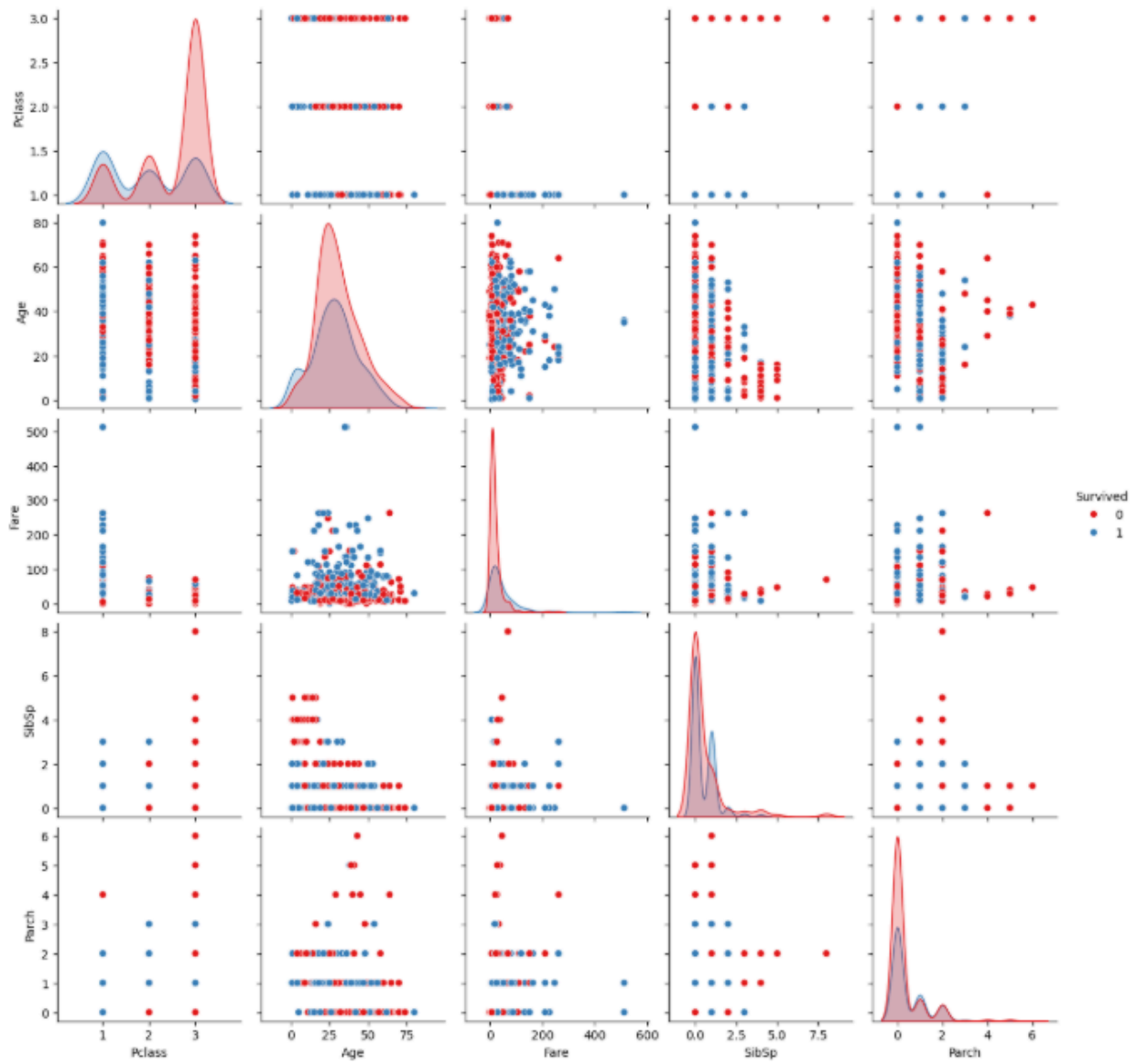
- 1)The age distribution is right-skewed. Most passengers are between 20 and 40 years old. There are also many children onboard. There are some missing values in the 'Age' column.
 - 2)Fare is heavily right-skewed, with most passengers paying under 100. A few outliers paid more than 500.
 - 3)Most passengers belonged to the 3rd class, followed by 1st and 2nd classes. Class 3 passengers were likely lower-income.
 - 4)There are significantly more male passengers than female passengers (~65% male).
 - 5)Median age of survivors is slightly lower than that of non-survivors. Young children had a higher survival chance.
 - 6)Most passengers boarded from port 'S' (Southampton), followed by 'C' (Cherbourg) and 'Q' (Queenstown).
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Observation:

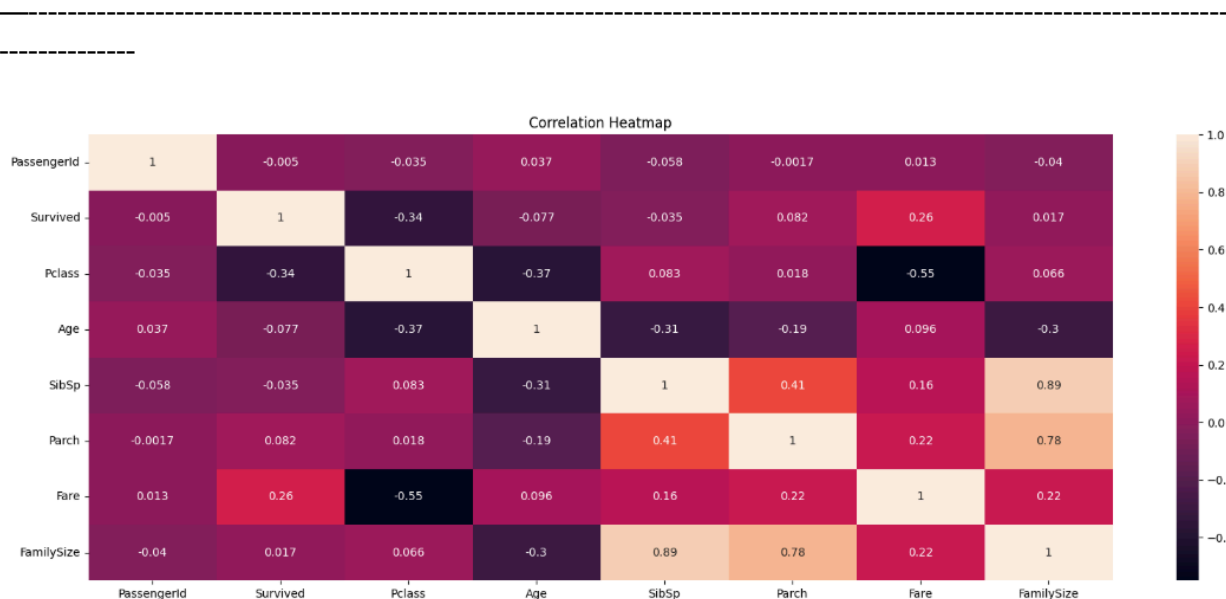
- Many high-fare, young/middle-aged passengers survived.
- Dense cluster of low-fare passengers didn't survive as often.

Pairplot of Selected Features



Observation:

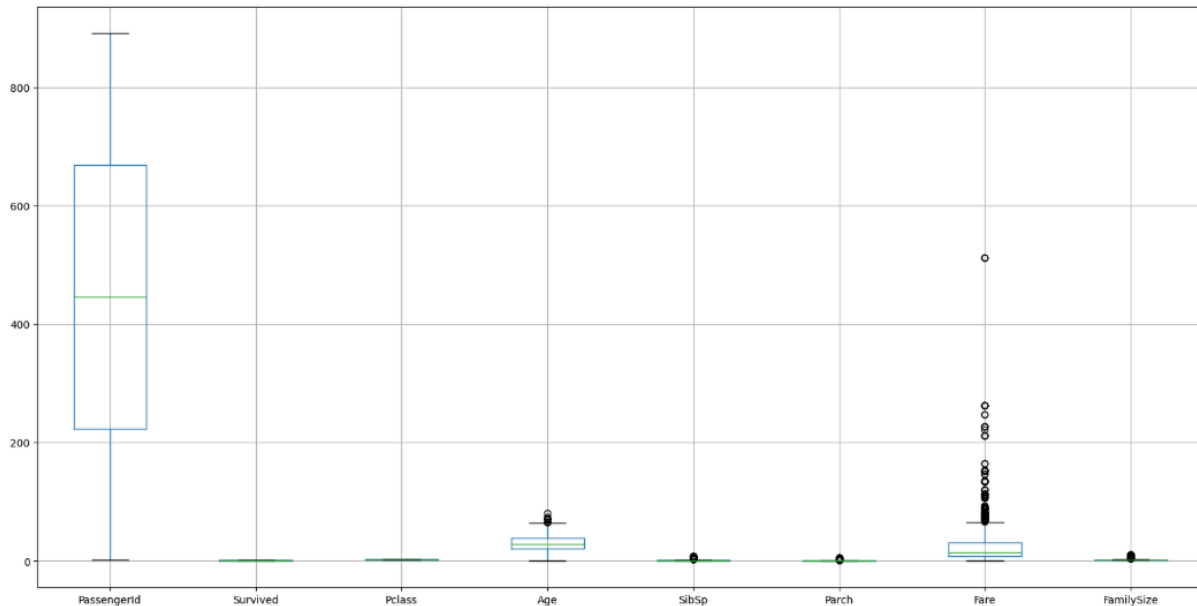
- **Survival vs. Pclass:**
 - Passengers in **1st class** had higher survival rates (many blue points).
 - Most passengers in **3rd class** did not survive (many red points).
- **Survival vs. Age:**
 - Survivors include a cluster of **younger passengers**, especially children.
 - Older age groups had a lower survival rate overall.
- **Survival vs. Fare:**
 - **Higher fares** are associated with higher survival chances.
 - Many low-fare passengers (likely from 3rd class) didn't survive.
- **Survival vs. SibSp & Parch:**
 - People with **1–2 siblings/spouses** or **1–3 parents/children** aboard had higher chances of survival.
 - Very high values of SibSp/Parch were rare and associated more with non-survivors.
- **Correlations:**
 - There's a visible **relationship between Pclass and Fare**: 1st class passengers paid higher fares.
 - **Fare and Age** also loosely trend together—older or wealthier passengers paid more.



Observation:

- Fare and Pclass are negatively correlated (higher class, higher fare).
- Survival is positively correlated with Fare and negatively with Pclass.

Box Plot



Observation:

- **Age** and **Fare** show significant spread and presence of outliers:
 - **Age:** Most passengers fall in the range of 20–40 years, with outliers extending beyond 60.
 - **Fare:** Highly right-skewed with extreme outliers. A few passengers paid over 500, while most fares are below 100.
- **SibSp**, **Parch**, and **FamilySize** have tight distributions near zero, indicating most passengers were either alone or with very few family members.
- **Pclass** and **Survived** have limited numerical spread since they are categorical in nature (ordinal/binary).
- **Passengerid** is a unique identifier and not useful for statistical analysis, but it's shown here for completeness.

Summary of Findings

- **Survival Rate** was ~38%.
- **Gender played a critical role** in survival: females had much higher survival rates.
- **Class mattered:** passengers in 1st class had significantly higher survival chances than those in 3rd class.
- **Children and young adults** were more likely to survive.
- **Fare paid** is positively correlated with survival: higher-paying passengers tended to survive.
- **Most passengers boarded at Southampton**, but survival rate varied slightly across ports.
- Some columns like **Cabin** had a lot of missing data, and may need further treatment or exclusion.
